

Midterm Exam 2 (Version 1)

CAS CS 132: Geometric Algorithms

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- ▷ You will have approximately 75 minutes to complete this exam. Make sure to read every question, some are easier than others.
- ▷ Do not remove any pages from the exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.
- ▷ We will not look at any work on the pages marked "*This page is intentionally left blank.*" You should use these pages for scratch work.

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1 Matrix Inverses

$$A = \begin{bmatrix} -2 & -2 & 7 \\ 1 & 2 & -1 \\ -2 & -3 & 5 \end{bmatrix}$$

A. (7 points) Determine the inverse of A , i.e., determine the matrix B such that $AB = BA = I$.

$$\begin{aligned} \begin{bmatrix} -2 & -2 & 7 & 1 & 0 & 0 \\ 1 & 2 & -1 & 0 & 1 & 0 \\ -2 & -3 & 5 & 0 & 0 & 1 \end{bmatrix} &\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 1 & 0 \\ -2 & -2 & 7 & 1 & 0 & 0 \\ -2 & -3 & 5 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 0 & 1 & 0 \\ 0 & 2 & 5 & 1 & 2 & 0 \\ 0 & 1 & 3 & 0 & 2 & 1 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 2 & 1 \\ 0 & 2 & 5 & 1 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 & -2 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & -1 & 3 & 2 \\ 0 & 1 & 0 & 3 & -4 & -5 \\ 0 & 0 & 1 & -1 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -7 & 11 & 12 \\ 0 & 1 & 0 & 3 & -4 & -5 \\ 0 & 0 & 1 & -1 & 2 & 2 \end{bmatrix} \end{aligned}$$

Solution.

$$\begin{bmatrix} -7 & 11 & 12 \\ 3 & -4 & -5 \\ -1 & 2 & 2 \end{bmatrix}$$

- B. (3 points) Use the matrix you determined in Part A to calculate the solution to the matrix equation $A\mathbf{x} = [1 \ 1 \ 1]^T$. For partial credit, determine the solution to the matrix equation $A\mathbf{x} = [1 \ 1 \ 1]^T$ using Gaussian elimination.

$$\vec{x} = A^{-1} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -7 & 11 & 12 \\ 3 & -4 & -5 \\ -1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -7 + 11 + 12 \\ 3 - 4 - 5 \\ -1 + 2 + 2 \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ -3 \end{bmatrix}$$

Solution.

$$\begin{bmatrix} 16 \\ -6 \\ -3 \end{bmatrix}$$

2 True/False

(10 points) Determine if each of the following statements is **True** or **False**. Bubble in your answers below.

- A. (1 point) For any stochastic matrix $A \in \mathbb{R}^{n \times n}$, if A has a unique steady-state distribution, then A is regular.

☐ True
☒ False

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- B. (1 point) If $\text{rank } A = 1$, then the matrix equation $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

☐ True
☒ False

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

- C. (1 point) If $\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\{\mathbf{v}_1, \mathbf{v}_3\}$ and $\{\mathbf{v}_2, \mathbf{v}_3\}$ are linearly independent, then so is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

☐ True
☒ False

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- D. (1 point) For matrices A, B, C in $\mathbb{R}^{n \times n}$, if $AB = AC$ then $B = C$.

☐ True
☒ False

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

- E. (1 point) For a subspace H of $\mathbb{R}^n$ and vector $\mathbf{v} \in \mathbb{R}^n$, if $\mathbf{v} \notin H$, then $\dim H < n$.

☒ True
☐ False

$$B = \{\mathbf{b}_1, \dots, \mathbf{b}_k\} \text{ is a basis for } H \quad B \cup \{\vec{r}\} \text{ is a basis for } K = \text{span}\{\mathbf{b}_1, \dots, \mathbf{b}_k, \vec{r}\}$$

$$\dim(H) < \dim(K) \leq n$$

- F. (1 point) For a matrix $A \in \mathbb{R}^{2026 \times 2027}$, it is possible for the rank of A to be the same as to $\dim(\text{Nul } A)$.

☐ True
☒ False

$$\text{rank}(A) + \dim(\text{Nul } A) = 2027$$

$$\text{rank}(A) = \dim(\text{Nul } A) \Rightarrow 2 \text{ rank } A = 2027 \Rightarrow \text{rank} = \frac{2027}{2}$$

- G. (1 point) If A is invertible, then both the L -part and the U -part of an LU -factorization of A are invertible.

☒ True
☐ False

- H. (1 point) For any stochastic matrix $A \in \mathbb{R}^{n \times n}$, if A is regular then $\dim(\text{Nul}(A - I)) = 1$.

☒ True
☐ False

$$\dim(\text{Nul}(A - I)) \geq 1 \quad \dim(\text{Nul}(A - I)) > 1 \Rightarrow A \text{ has multiple steady-state distributions.}$$

- I. (1 point) For matrices A and B in $\mathbb{R}^{m \times n}$, let $\begin{bmatrix} A \\ B \end{bmatrix}$ denote the $\mathbb{R}^{2m \times n}$ obtained by stacking A and B vertically.

For any vector $\mathbf{b} \in \mathbb{R}^n$, if $A\mathbf{x} = \mathbf{b}$ is consistent and $B\mathbf{x} = \mathbf{b}$ is consistent then so is $\begin{bmatrix} A \\ B \end{bmatrix} \mathbf{x} = \begin{bmatrix} \mathbf{b} \\ \mathbf{b} \end{bmatrix}$.

☐ True
☒ False

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ is inconsistent}$$

- J. (1 point) Given linear transformations $S: \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, if $m < n$, then it is possible for $T \circ S$ to be invertible (recall that $(T \circ S)(\mathbf{v}) = T(S(\mathbf{v}))$).

☒ True
☐ False

$$S(\vec{v}) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \vec{v} \quad T(\vec{v}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \vec{v}$$

3 Subspaces

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \begin{bmatrix} 2x + 5y \\ -x - 2y - z \\ x + 3y - z \end{bmatrix}$$

- A. (2 points) Determine the matrix A that implements the above transformation, i.e., determine the matrix A such that the above transformation is equivalent to $\mathbf{x} \mapsto A\mathbf{x}$.

Solution.

$$\begin{bmatrix} 2 & 5 & 0 \\ -1 & -2 & -1 \\ 1 & 3 & -1 \end{bmatrix}$$

- B. (4 points) Determine a basis for $\text{Nul } A$.

$$\sim \begin{bmatrix} 1 & 3 & -1 \\ -1 & -2 & -1 \\ 2 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = -5x_3$
 $x_2 = 2x_3$
 x_3 is free

$$x_3 \begin{bmatrix} -5 \\ 2 \\ 1 \end{bmatrix}$$

Solution.

$$\text{Nul}(A) = \text{span} \left\{ \begin{bmatrix} -5 \\ 2 \\ 1 \end{bmatrix} \right\}$$

You may also use this page to show your work for both Part B and Part C if necessary.

C. (2 points) Determine a basis for $\text{Col } A$.

Solution.

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix} \right\}$$

D. (2 points) Let T denote the transformation from Part A and let H be the subspace

$$\{\mathbf{v} : \mathbf{v} \in \text{ran } T \text{ and } \mathbf{e}_3^T \mathbf{v} = 0\}$$

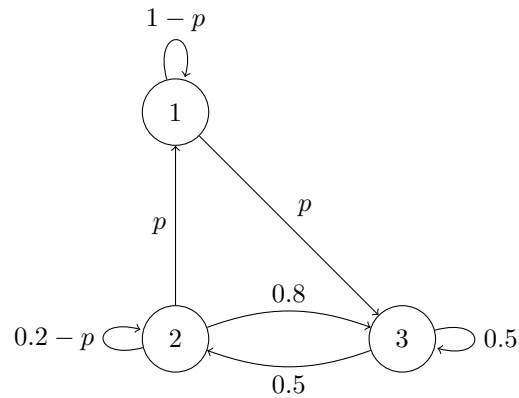
That is, H is **the subspace of vectors in the range of T whose third components are 0**. Note that H is a subspace of $\text{ran } T$, which is a subspace of $\mathbb{R}^3$. Determine a basis for H .

$$\begin{array}{lll} x + 3y - z = 0 & \begin{array}{l} x = 1 \\ y = 0 \\ z = 1 \end{array} & \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix} \end{array}$$

Solution.

$$H = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$$

4 Markov Chains



- A. (2 points) Determine the transition matrix A of the above state diagram in terms of p . The transitions out of state 1 should appear in the first column of A , out of 2 in the second column, and out of 3 in the third column.

Solution.

$$\begin{bmatrix} 1-p & p & 0 \\ 0 & 0.2-p & 0.5 \\ p & 0.8 & 0.5 \end{bmatrix}$$

- B. (2 points) Is the matrix A regular in the case that $p = 0$? Justify your answer.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.2 & 0.5 \\ 0 & 0.8 & 0.5 \end{bmatrix}$$

Solution. ☐ Yes ☒ No

Justification.

$$A \vec{e}_1 = \vec{e}_1 \Rightarrow A^k \vec{e}_1 = \vec{e}_1, \text{ first col. of } A^k \text{ has zeros}$$

C. (4 points) Determine $\lim_{k \rightarrow \infty} A^k [0.2 \ 0.4 \ 0.4]^T$ in the case that $p = 0.1$.

$$A - I = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0 & 0.1 & 0.5 \\ 0.1 & 0.8 & 0.5 \end{bmatrix} - I = \begin{bmatrix} -0.1 & 0.1 & 0 \\ 0 & -0.9 & 0.5 \\ 0.1 & 0.8 & -0.5 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 9 & -5 \\ 1^{-1} & 8^{+1} & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 9 & -5 \\ 0 & 9 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0^{+1} & -5/9 \\ 0 & 1 & -5/9 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -5/9 \\ 0 & 1 & -5/9 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 5/9 x_3$$

$$x_2 = 5/9 x_3$$

x_3 is free

$$x_1 + x_2 + x_3 = \frac{5}{9} x_3 + \frac{5}{9} x_3 + \frac{9}{9} x_3 = \frac{19}{9} x_3 = 1$$

$$x_1 = \frac{5}{19}$$

$$x_2 = 5/19$$

$$x_3 = 9/19$$

Solution.

$$\begin{bmatrix} 5/19 \\ 5/19 \\ 9/19 \end{bmatrix}$$

- D. (2 points) Determine $\lim_{k \rightarrow \infty} A^k [0.2 \ 0.4 \ 0.4]^T$ in the case that $p = 0$. For partial credit, determine a general form solution that describes the steady-state vectors of A in the case that $p = 0$ (the steady state vectors of a matrix A are the vectors $\mathbf{v}$ such that $A\mathbf{v} = \mathbf{v}$).

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.2 & 0.5 \\ 0 & 0.8 & 0.5 \end{bmatrix}$$

$$A - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -0.8 & 0.5 \\ 0 & 0.8 & -0.5 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & -5/8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

x_1 is free

$$x_2 = 5/8 x_3$$

x_3 is free

$$5/8 x_3 + 8/8 x_3 = 1$$

$$13/8 x_3 = 1$$

$$x_3 = \frac{8}{13} \cdot \frac{4}{5} = 32/65$$

$$x_2 = \frac{5}{13} \cdot \frac{4}{5} = 20/65$$

$$x_1 = \frac{1}{5} = 13/65$$

Solution.

$$\begin{bmatrix} 13/65 \\ 20/65 \\ 32/65 \end{bmatrix}$$

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